

Data Provisioning

Data Provisioning is the process of collecting, integrating, transforming, and preparing data from multiple sources for analysis, reporting, and business intelligence applications. It ensures that accurate and useful data is available to users when required.

Organizations generate large amounts of data from: Databases, Websites

Raw data is often incomplete and inconsistent.

Objectives of Data Provisioning

1. Collect data from multiple sources
2. Improve data quality
3. Transform data into usable format
4. Provide accurate information
5. Support analytics and reporting

Data Warehouse

is a centralized repository that stores integrated historical data from multiple sources for reporting and analysis. It stores cleaned and structured data.

A data warehouse supports: Business Intelligence, Analytics, Reporting

Characteristics of Data Warehouse

1. Subject-oriented
2. Integrated
3. Time-variant

Advantages

1. Centralized data storage.
2. Better business analysis.

Disadvantages

1. High implementation cost.
2. Complex maintenance.

Schemas in Data Warehouse

A Schema is the logical structure that defines how data is organized in a data warehouse.

Types of Schemas

1. Star Schema
2. Snowflake Schema
3. Fact Constellation Schema

A. Star Schema - contains one central fact table connected to multiple dimension tables.

Structure

Dimension Tables

↓

Fact Table

↑

Dimension Tables

Advantages

1. Simple design.
2. Faster query execution.

Disadvantages

1. Data redundancy.

B. Snowflake Schema

Snowflake Schema is an extension of star schema where dimension tables are normalized.

Advantages

1. Reduced redundancy.
2. Better data organization.

Disadvantages

1. Complex queries.

Data Quality

Data Quality refers to the accuracy, completeness, consistency, and reliability of data.

Poor quality data leads to: Incorrect analysis, Wrong decisions, Business losses

Characteristics of Data Quality - Accuracy, Completeness, Consistency, Validity, Timeliness

Advantages

1. Better decision-making.
2. Improved analytics accuracy.

Challenges

1. Missing data.
2. Duplicate records.

Data Profiling

Data Profiling is the process of analyzing data to understand its structure, quality, and content.

Objectives - Identify missing values, Detect duplicates, Analyze patterns, Check consistency

Types of Data Profiling

A. Structure Profiling - Analyzes data format

B. Content Profiling - Checks actual values.

C. Relationship Profiling - Analyzes relationships between datasets.

Advantages

1. Improves data quality.
2. Detects anomalies.
3. Helps preprocessing.

Applications - Data cleaning, ETL systems

Data Enrichment

Data Enrichment is the process of enhancing existing data by adding additional useful information. Raw data may not contain sufficient details. Additional information improves analysis quality.

Advantages

1. Better insights.
2. Improved analytics.

Disadvantages

1. Increased storage requirements.
2. Additional processing cost.

Data Duplication

Data Duplication occurs when identical or repeated data records exist in multiple locations.

Causes

- Multiple data sources
- Improper integration
- System synchronization errors

Duplicate Detection Techniques

Data matching, Hashing, AI-based duplicate detection

Positive Impacts of Data Duplication

1. Faster Query Performance

Frequently accessed data can be duplicated to reduce query processing time.

2. Improved Data Availability

Duplicate copies ensure data remains available even if one source fails.

3. Better Backup and Recovery

Multiple copies of data help in disaster recovery and fault tolerance.

4. Supports Distributed Access

Users from different locations can access local copies of data more efficiently.

Negative Impacts of Data Duplication

1. Increased Storage Cost

Duplicate data consumes additional storage space.

2. Data Inconsistency

Different copies may contain different values if updates are not synchronized.

3. Increased Maintenance Effort

Managing and updating multiple copies of the same data becomes difficult.

4. Slower ETL Process - Extra time is required to load, update, and synchronize duplicate records.

5. Reduced Data Quality

Duplicate records can lead to inaccurate reports and analysis.

6. Increased Processing Overhead

Additional resources are required for data management and synchronization.

Impact on Data Warehouse Efficiency

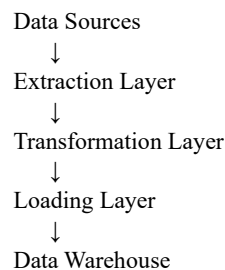
- May improve query response time.
- Increases storage and maintenance costs.
- Can reduce data quality and consistency.
- Affects ETL performance and resource utilization.

ETL Architecture

ETL Architecture defines the framework used for Extracting, Transforming, and Loading data into a data warehouse.

ETL Full Form = Extract → Transform → Load

ETL Architecture Workflow



Components of ETL Architecture

- 1. Source Layer** - Collects raw data.
- 2. Staging Area** - Temporary storage for processing.
- 3. Transformation Layer** - Cleans and converts data.
- 4. Target Layer** - Stores processed data.

Advantages

1. Structured data integration.
2. Better analytics support.

Disadvantages

1. Complex implementation.
2. Time-consuming processing.

Extraction Concept

Extraction is the process of retrieving data from various data sources for further processing.

Types of Extraction

A. Full Extraction - Entire dataset extracted.

B. Incremental Extraction - Only changed data extracted.

Advantages

1. Data integration support.
2. Faster analytics preparation.

Applications

- ETL systems

- Data migration

Change Data Capture (CDC)

CDC is a technique used to identify and capture changes made to database records.

CDC detects: Insertions, Updates, Deletions

without processing the entire dataset.

Working

Database Changes



Capture Modified Records



Update Data Warehouse

Advantages

1. Faster data synchronization.
2. Reduced processing overhead.

Disadvantages

1. Complex implementation.
2. Additional monitoring required.

Applications

Real-time BI, Database replication, Streaming systems

Transformation Concept

Transformation is the process of converting raw extracted data into a clean, structured, and suitable format before loading it into a data warehouse.

Data collected from different sources may contain: Missing values, Duplicate records, Different formats, Errors and inconsistencies

Transformation improves data quality and prepares data for analytics.

Transformation Operations

A. Data Cleaning - Removes errors and duplicate records.

B. Data Filtering - Selects required data only.

C. Data Aggregation - Combines multiple records into summarized form.

D. Data Normalization - Converts data into a standard format.

E. Data Conversion - Changes data types or formats.

Advantages

1. Improves data quality.
2. Increases consistency.

Disadvantages

1. Time-consuming process.
2. Complex for large datasets.

Lookups

Lookup is a transformation technique used to retrieve related information from another table or dataset during ETL processing. Lookups help combine and enrich data using reference tables.

Working

Step 1: Input data received.

Step 2: Lookup table searched.

Step 3: Matching value retrieved.

Example

Employee table retrieves department name using department ID.

Role of Lookups in ETL

1. Data Enrichment

Lookups add additional information to records by fetching data from reference tables.

2. Data Validation

Used to verify whether incoming data exists in a master or reference table.

3. Data Standardization

Ensures data values are consistent with predefined standards.

4. Key Generation

Helps obtain surrogate keys from dimension tables in data warehouses.

5. Data Mapping

Maps source data values to corresponding target values.

6. Duplicate Detection

Identifies existing records and prevents duplicate entries.

Types of Lookups

A. Connected Lookup - Connected directly to ETL pipeline.

B. Unconnected Lookup - Called only when required.

Time Lag

is the delay between data generation and data availability in the target system or data warehouse.

Causes of Time Lag - Network delays, Large data processing, Batch processing, Slow ETL execution

Example: Sales data generated at 10 AM reaches warehouse at 11 AM.

Effects of Time Lag - Delayed analytics, Outdated reports, Slower decision-making

Methods to Reduce Time Lag - Real-time ETL, Incremental loading, Faster processing systems

Advantages of Reducing Time Lag - Faster reporting, Real-time analytics, Better business decisions.

Formats

Formats define the structure and representation of data in databases and ETL systems.

Common Data Formats

Format Description

CSV	Comma-separated values
JSON	JavaScript Object Notation
XML	Extensible Markup Language
Parquet	Column-based storage format
Excel	Spreadsheet format

Importance of Formats - Ensures compatibility, Simplifies data exchange, Improves processing efficiency.

Challenges - Format incompatibility, Data conversion complexity.

Consistency

Consistency means maintaining accurate and uniform data across all systems and databases.

Inconsistent data causes: Incorrect reports, Duplicate information, Wrong decisions

Example - Customer name stored differently in two systems.

Types of Consistency

A. Data Consistency - Same data values across systems.

B. Transaction Consistency - Correct execution of transactions.

Methods to Maintain Consistency - Data validation, Synchronization, Duplicate removal

Advantages - Accurate analytics, Reliable reporting.

Loading Concept

Loading is the process of storing transformed data into the target data warehouse or database. It is the final step of the ETL process.

Loading Workflow

Extract Data -> Transform Data -> Load into Warehouse

Types of Loading

1. Initial Loading
2. Incremental Loading

Advantages

1. Centralized storage.
2. Supports reporting and analytics.

Initial Loading - loads the complete dataset into the target system for the first time.

Features: Full dataset transferred, Performed once initially

Advantages

1. Complete data availability.
2. Simple implementation.

Disadvantages

1. Time-consuming.
2. High resource usage.

Incremental Loading

Incremental Loading transfers only newly added or modified records instead of loading the entire dataset again. Used after initial loading to improve efficiency.

Working: Detect Changed Data -> Load Only Modified Records

Advantages

1. Faster processing.
2. Reduced storage usage.
3. Efficient updates.

Disadvantages

1. Complex change tracking.

Late Arriving Facts

are data records that arrive after the expected processing time.

Sometimes data is delayed because of: Network failures, System delays, Processing errors

Problems Caused: Incorrect reports, Delayed analytics, Inconsistent data.

Handling Techniques

- Data reprocessing
- Timestamp management
- Incremental updates

Advantages of Proper Handling

1. Accurate reporting.
2. Improved consistency.

What is Staging?

Staging is a temporary storage area used in ETL systems before data is transformed and loaded into the warehouse.

Raw extracted data is stored in staging area for: Cleaning, Validation, Transformation

Workflow: Source Data->Staging Area->Transformation->Data Warehouse

Advantages

1. Improves ETL management.
2. Supports error handling.
3. Simplifies transformation.

Disadvantages: Requires additional storage.

Data Marts

A Data Mart is a smaller, department-specific subset of a data warehouse. Data marts focus on a particular business function such as: Sales, Finance, Marketing

Types of Data Marts

1. Dependent DM - Derived from data warehouse.
2. Independent DM - Created from operational systems.
3. Hybrid DM - Combines multiple data sources.

Sales Data Mart

Product ID Product Name Region Sales Amount

P101	Laptop	North	50000
P102	Mobile	South	30000
P103	Tablet	East	20000
P104	Laptop	West	45000

Purpose of a Data Mart in BI System

1. Department-Specific Data

Stores data related to a particular business function such as sales, finance, or marketing.

2. Faster Query Processing

Contains a smaller subset of data, resulting in quicker report generation.

3. Better Decision Making

Provides relevant information for managers and business analysts.

4. Improved Performance

Reduces the load on the central data warehouse.

5. Easy Data Access

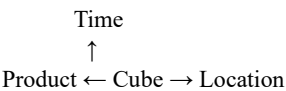
Users can easily access and analyze department-specific data.

Cubes

A DC is a multidimensional data structure used for fast analytical processing in Business Intelligence systems. Data cubes organize data across multiple dimensions.

Example Dimensions - Time, Product, Location

Cube Structure



Operations on Cubes - Slice, Dice, Drill-down, Roll-up

Advantages

- 1. Fast analytical queries.
- 2. Multidimensional analysis.

Disadvantages

- 1. Large storage requirement.
- 2. Complex maintenance.

Data Visualization

is the graphical representation of data and information using charts, graphs, dashboards, and visual elements to make analysis easier and more understandable. Raw data is difficult to understand directly.

Visualization helps: Identify patterns, Compare performance, Detect trends, Support business decisions

Objectives of Data Visualization

- 1. Simplify complex data
- 2. Improve understanding
- 3. Support decision-making
- 4. Detect trends and anomalies
- 5. Enhance reporting efficiency

What is a Business Report?

A Business Report is a structured document that presents business data, analysis, insights, and recommendations for decision-making.

Business reports help organizations monitor: Sales, Finance, Customer behavior, Inventory

Reports are generated using BI tools and analytics systems.

Features of Business Reports - Organized information, Visual representation, Performance analysis, Decision support

Types of Business Reports

Type	Purpose
Operational Report	Daily operations monitoring
Financial Report	Financial analysis
Analytical Report	Data analysis and trends
Performance Report	KPI monitoring

Components of Business Reporting Systems

Business Reporting Systems are systems that collect, process, analyze, and present business information.

Main Components

- A. Data Sources - Collect raw business data from databases, applications, and external systems.
- B. Data Warehouse - Stores integrated historical data.
- C. ETL Process - Extracts, transforms, and loads data.
- D. Analytics Engine - Processes data and generates insights.
- E. Reporting Tools - Generate charts, dashboards, and reports.
- F. Visualization Layer-Displays information graphically.

Advantages

- 1. Centralized reporting.
- 2. Faster business analysis.
- 3. Improved monitoring.

Data and Information Visualization

Data Visualization represents raw data graphically, while Information Visualization presents processed information and insights visually.

Difference Between Data and Information Visualization

Parameter Data Visualization Information Visualization

Focus	Raw data	Processed insights
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Parameter Data Visualization Information Visualization

Purpose Data representation Knowledge presentation

Example Sales chart Business performance dashboard

Types of Charts and Graphs

A. Bar Chart - Represents data using rectangular bars.

Applications - Comparing sales, Product analysis

B. Line Graph - Represents data trends over time.

Applications - Stock market trends

C. Pie Chart - Represents data as portions of a circle.

Applications - Market share analysis

D. Histogram - Shows frequency distribution of data.

Applications - Statistical analysis

E. Scatter Plot - Displays relationships between two variables.(ml)

Visual Analytics

Visual Analytics combines data visualization, analytics, and interactive exploration to analyze large datasets.

Visual analytics helps users: Explore patterns, Detect anomalies, Analyze trends, Make faster decisions

Components of Visual Analytics - Data processing, Visualization tools, Interactive analysis, AI and analytics

Advantages - Faster insights, Interactive exploration, Better decision-making.

Disadvantages - High computational requirements, Complex implementation.

Performance Dashboards

is a visual interface that displays important business metrics and KPIs in real time.

Dashboards summarize business performance using: Charts, Graphs, Indicators, Tables

Types of Dashboards

A. Operational Dashboard -Monitors daily operations.

B. Strategic Dashboard - Supports long-term planning.

C. Analytical Dashboard - Provides detailed analytical insights.

Dashboard Components

1. KPIs
2. Charts and graphs
3. Alerts and notifications
4. Reports

Advantages

1. Real-time monitoring.
2. Improved decision-making.
3. Better performance tracking.

Disadvantages

1. Complex design.
2. Information overload possible.

Business Performance Management (BPM)

is the process of monitoring, analyzing, and improving organizational performance using BI tools and analytics.

Objectives of BPM - Improve efficiency, Track business goals, Optimize performance, Support strategic planning

BPM Process: Business Goals->Data Collection->Performance Analysis->Decision Making->Performance Improvement

Components of BPM

1. KPI monitoring
2. Reporting systems
3. Forecasting
4. Performance dashboards

Advantages

1. Better business control.
2. Improved productivity.
3. Goal tracking.

Disadvantages

1. Expensive implementation.
2. Requires quality data.

Introduction to BI Tools

BI Tools are software applications used to collect, process, analyze, visualize, and report business data for decision-making. Organizations generate huge volumes of data daily.

Objectives of BI Tools

1. Simplify data analysis
2. Generate business reports
3. Support data visualization
4. Improve decision-making
5. Enable real-time monitoring

Common Features of BI Tools

Data connectivity, Dashboard creation, Interactive visualization, Reporting and analytics, Data sharing and collaboration

1. Tableau

is a powerful data visualization and analytics tool used for creating interactive dashboards and reports. Tableau helps users analyze large datasets visually without extensive programming knowledge.

It supports:

- Drag-and-drop interface
- Real-time analytics
- Interactive dashboards

Features of Tableau

1. Interactive dashboards
2. Real-time data analysis
3. Drag-and-drop visualization
4. Data blending and integration

Advantages

1. Easy to use.
2. Excellent visualization capabilities.

Disadvantages

1. Expensive licensing.
2. Limited data preprocessing.

2. Power BI

tool developed by Microsoft for data visualization, reporting, and business analytics. Power BI integrates easily with Microsoft products and cloud services.

It supports:

- Real-time dashboards
- AI-powered analytics
- Data sharing

Components of Power BI

1. Power BI Desktop
2. Power BI Service
3. Power BI Mobile

Features - Interactive dashboards, Cloud integration, AI-based insights, Real-time monitoring, Data connectivity

Advantages

1. Easy Microsoft integration.
2. Affordable pricing.

Disadvantages

1. Limited customization compared to Tableau.

Dundas BI

is a Business Intelligence platform used for reporting, dashboard creation, and data visualization. Dundas BI provides flexible and customizable analytics solutions.

Features - Interactive dashboards, Advanced charting, Data integration, Custom analytics

Advantages

1. Highly customizable.
2. Strong dashboard features.

Disadvantages

1. Complex for beginners.
2. Higher implementation cost.

4. Oracle BI

is an enterprise Business Intelligence platform used for analytics, reporting, and performance management. Oracle BI supports large-scale enterprise analytics and integrates with Oracle databases and cloud systems.

Features - Enterprise reporting, Real-time analytics, KPI dashboards, Predictive analytics, Strong database integration

Advantages

1. Highly scalable.
2. Strong enterprise support.
3. Advanced analytics features.

Disadvantages

1. Expensive.
2. Complex implementation.

5. MS Excel

MS Excel is a spreadsheet application used for data analysis, calculations, visualization, and reporting. Excel is one of the most commonly used BI tools for small and medium-scale analytics.

Features - Spreadsheet calculations, Charts and graphs, Pivot tables, Data filtering and sorting, Basic analytics

Comparison of BI Tools

Tool	Main Strength	Best Use
Tableau	Advanced visualization	Interactive dashboards
Power BI	Microsoft integration	Enterprise analytics
Dundas BI	Custom dashboards	Enterprise reporting
Oracle BI	Large-scale analytics	Enterprise management
MS Excel	Simplicity	Small-scale analysis